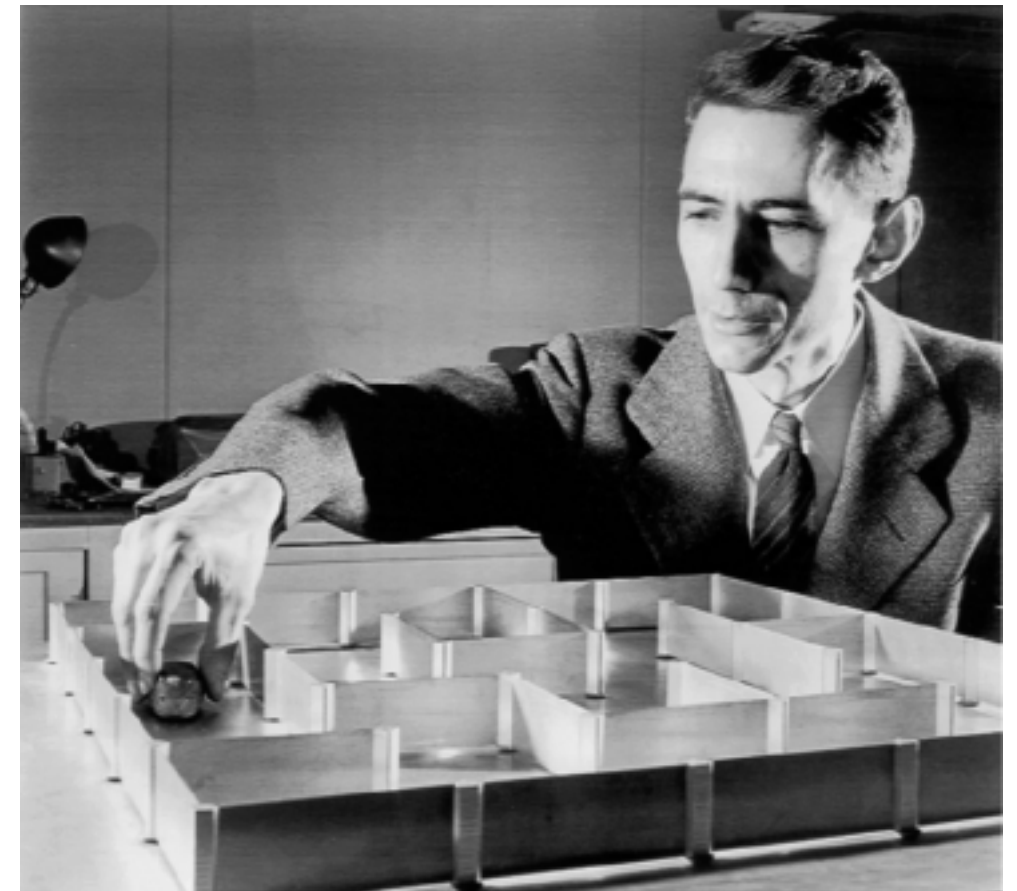


Class 13: Information Theory

- **Today: introduction to information theory**
 - Morse code analysis**
 - information content of a message**
- **Thursday: determining authorship**
- **Lab: Analyzing information content of text from different sources**

Claude Shannon (1916–2001)

- **Born in Petoskey, Michigan**
- **Completed two undergraduate degrees at the University of Michigan**
- **Master's thesis in electrical engineering on binary arithmetic revolutionized electrical engineering and formed the basic of digital circuit design**
- **PhD thesis: “An Algebra for Theoretical Genetics”**
- **Involved in code breaking during World War 2**



Shannon (+ robotic mouse)

Shannon's paper

The Bell System Technical Journal

Vol. XXVII

July, 1948

No. 3

A Mathematical Theory of Communication

By C. E. SHANNON

Mathematical theory of communication

The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point. Frequently the messages have *meaning*; that is they refer to or are correlated according to some system with certain physical or conceptual entities. These semantic aspects of communication are irrelevant to the engineering problem. The significant aspect is that the actual message is one *selected from a set* of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design.

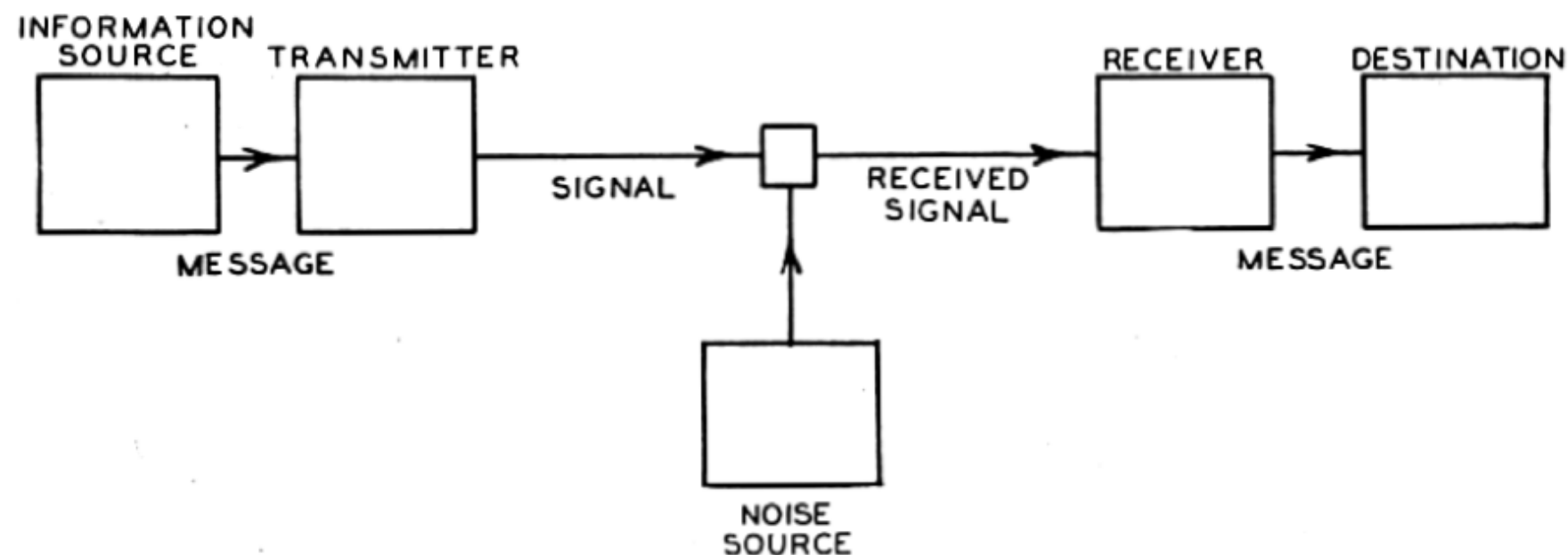


Fig. 1—Schematic diagram of a general communication system.

Mathematical theory of communication

Message must be *encoded*,
then *transmitted*,
then *decoded*.

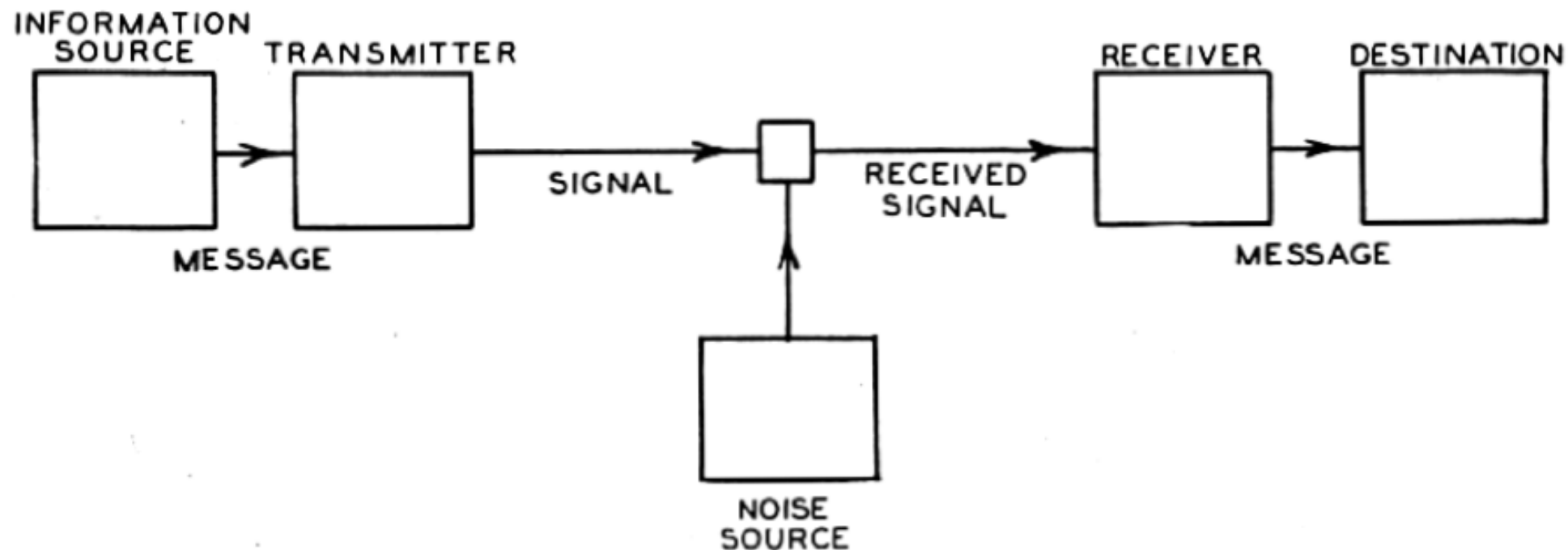


Fig. 1—Schematic diagram of a general communication system.

Forward problem

Given a message, can we code and transmit the message?

Mathematical theory of communication

Message must be *encoded*,
then *transmitted*,
then *decoded*.

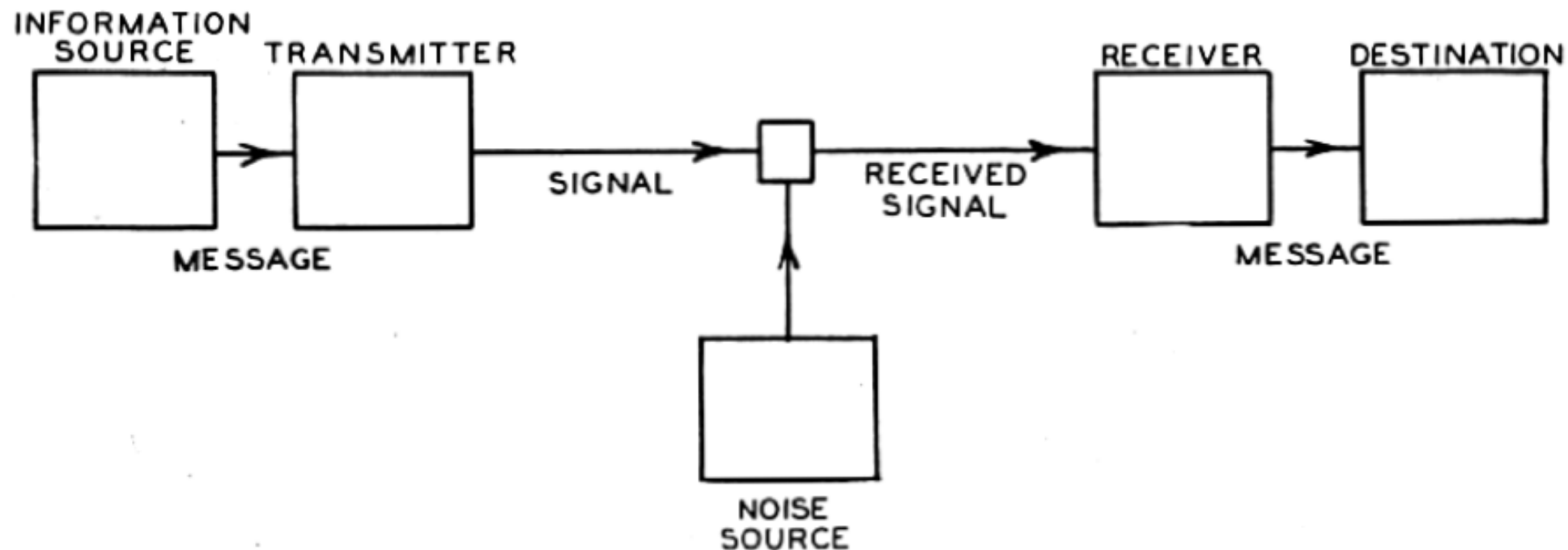


Fig. 1—Schematic diagram of a general communication system.

Inverse problem

Given the received signal, can we decode the message?

Messages take many forms

- **There are many ways to communicate**
- **Messages take many different forms**

Audio

- **English**
- **French**
- **Mandarin**
- **Arabic**

Visual

- **Sign language**
- **...**

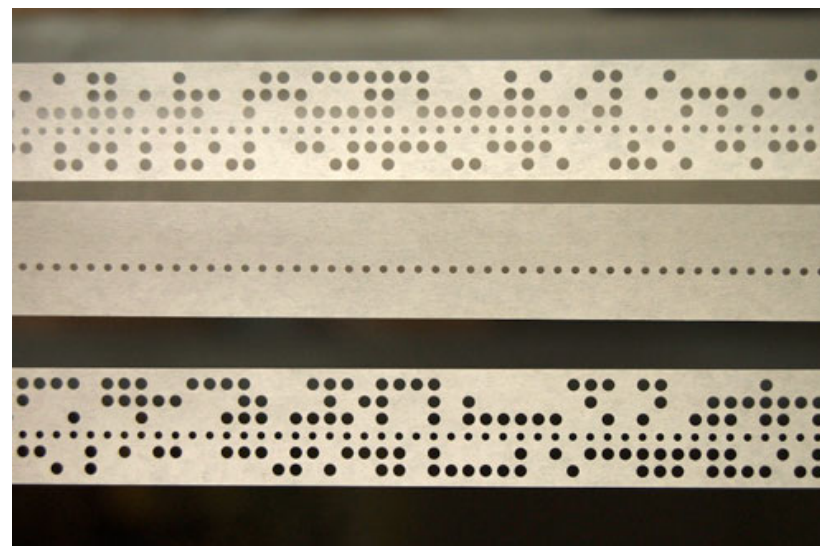
Electronic

- **...**

Teletype



- Originated in early twentieth century for sending messages of text electronically
- Could also be used to archive text



USTTY			
hex	binary	LTRS	FIGS
0x00	00000	BLANK	BLANK
0x01	00001	E	3
0x02	00010	LF	LF
0x03	00011	A	-
0x04	00100	SP	SP
0x05	00101	S	BELL
0x06	00110	I	8
0x07	00111	U	7
0x08	01000	CR	CR
0x09	01001	D	\$
0x0A	01010	R	4
0x0B	01011	J	'
0x0C	01100	N	,
0x0D	01101	F	!
0x0E	01110	C	:
0x0F	01111	K	(
0x10	10000	T	5
0x11	10001	Z	"
0x12	10010	L	)
0x13	10011	W	2
0x14	10100	H	#
0x15	10101	Y	6
0x16	10110	P	0
0x17	10111	Q	1
0x18	11000	O	9
0x19	11001	B	?
0x1A	11010	G	&
0x1B	11011	FIGS	FIGS
0x1C	11100	M	.
0x1D	11101	X	/
0x1E	11110	V	;
0x1F	11111	LTRS	LTRS

Morse code

- Developed in 1836 by artist Samuel Morse, physicist Joseph Henry, and inventor Alfred Vail for sending messages on a telegraph system
- Still in use today
- Experienced users can achieve speeds of 40 words per minute



International Morse Code

1. The length of a dot is one unit.
2. A dash is three units.
3. The space between parts of the same letter is one unit.
4. The space between letters is three units.
5. The space between words is seven units.

A	• —
B	— • • •
C	— • — •
D	— • •
E	•
F	• • — •
G	— — •
H	• • • •
I	• •
J	• — — —
K	— • —
L	• — • •
M	— —
N	— •
O	— — —
P	• — — •
Q	— — • —
R	• — •
S	• • •
T	—

U	• • —
V	• • • —
W	• — —
X	— • • —
Y	— • — —
Z	— — • •

1	• — — — —
2	• • — — —
3	• • • — —
4	• • • • —
5	• • • • •
6	— • • • •
7	— — • • •
8	— — — • •
9	— — — — •
0	— — — — —

Morse code design: letter frequencies

Letter frequencies

E 11.28%, T 8.46%, A 7.52%, I 7.52%, N 7.52%, O 7.52%,
H 6.02%, R 5.83%, D 4.14%, L 3.76%, U 3.20%, C 2.82%,
M 2.82%, F 2.35%, W 1.88%, Y 1.88%, G 1.60%, P 1.60%,
B 1.50%, V 1.13%, K 0.75%, Q 0.47%, J 0.38%, X 0.38%,
Z 0.19%

Most common letters have short codes

E	T	A	I	N	O	H
•	—	•—	••	—•	— — —	••••

Code design: timing specifics

- **Dot = line closure for one unit of time and then line open for one unit of time**
- **Dash = line closure for three units of time and line open for one unit of time**
- **Letter Space = line open for three units of time**
- **Word Space = line open for six units of time**
- **Additional Rule: cannot have two successive spaces**

Possible signals

- **Dot = 2 time units**
- **Dash = 4 time units**
- **Dot-space = 5 time units**
- **Dash-space = 7 time units**
- **Dot-wordspace = 8 time units**
- **Dash-wordspace = 10 time units**

Total number of signals transmitted in unit time

Let $N(t)$ be the number of signals that can form in time t

$$\begin{aligned} N(t) = & \text{number of signals ending in a “dot”} \\ & + \text{number of signals ending in a “dash”} \\ & + \text{number of signals ending in a “dash-space”} \\ & + \text{number of signals ending in a “dot-space”} \\ & + \text{number of signals ending in a “dot-wordspace”} \\ & + \text{number of signals ending in a “dash-wordspace”} \end{aligned}$$

Total number of signals transmitted in unit time

$$\begin{aligned} N(t) = & \text{number of signals ending in a "dot"} \\ & + \text{number of signals ending in a "dash"} \\ & + \text{number of signals ending in a "dash-space"} \\ & + \text{number of signals ending in a "dot-space"} \\ & + \text{number of signals ending in a "dot-wordspace"} \\ & + \text{number of signals ending in a "dash-wordspace"} \end{aligned}$$

$$\begin{aligned} N(t) = & N(t - 2) + N(t - 4) + N(t - 5) + N(t - 7) \\ & + N(t - 8) + N(t - 10) \end{aligned} \quad \text{"Difference equation"}$$

Aside: another difference equation (Fibonacci sequence)

- Sequence of numbers where each is the sum of the previous two

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

- Let $F(n)$ be the n th Fibonacci number



Fibonacci
(c. 1170 – c. 1250)

Come up with the difference equation for $F(n)$

Aside: another difference equation (Fibonacci sequence)

- Sequence of numbers where each is the sum of the previous two

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, ...

- Let $F(n)$ be the n th Fibonacci number

$$F(1) = 0, \quad F(2) = 1,$$

$$F(n) = F(n - 1) + F(n - 2), \quad n \geq 3$$

Aside: another difference equation (Fibonacci sequence)

- **How do we solve this equation?**

$$F(1) = 0, \quad F(2) = 1,$$

$$F(n) = F(n-1) + F(n-2), \quad n \geq 3$$

- **Try $F(n) = x^n$ for some unknown x**

Aside: Fibonacci difference equation

$$F(n) = F(n-1) + F(n-2) \quad \Rightarrow \quad x^n = x^{n-1} + x^{n-2}$$

- Hence x satisfies a quadratic

$$x^2 - x - 1 = 0$$

- Two solutions

$$x = \frac{1 + \sqrt{5}}{2}, \frac{1 - \sqrt{5}}{2}$$

- Define $\phi = \frac{1 + \sqrt{5}}{2}$ (the Golden ratio), and then

$$x = \phi, -\phi^{-1}$$

Aside: Fibonacci difference equation

- **General solution will be a "linear combination"**

$$F(n) = A \phi^n + B(-\phi^{-1})^n = A \phi^n + C \phi^{-n}$$

- **At large n ,**

$$F(n) \approx a \phi^n = a x^n$$

Returning to the problem

$$N(t) = N(t - 2) + N(t - 4) + N(t - 5) + N(t - 7) \\ + N(t - 8) + N(t - 10)$$

- **Try solution of the form $N(t) = x^t$**

Returning to the problem

$$N(t) = N(t - 2) + N(t - 4) + N(t - 5) + N(t - 7) \\ + N(t - 8) + N(t - 10)$$

- **Try solution of the form $N(t) = x^t$**

$$1 = x^{-2} + x^{-4} + x^{-7} + x^{-8} + x^{-10}$$

- **Writing $x = \frac{1}{\mu}$ gives**

$$0 = f(\mu) = \mu^{10} + \mu^8 + \mu^7 + \mu^5 + \mu^4 + \mu^2 - 1$$

- **Largest value of x and smallest value of μ**

$$x \approx 1.453, \quad \mu = 0.688$$

Express rate in binary

The choice of a logarithmic base corresponds to the choice of a unit for measuring information. If the base 2 is used the resulting units may be called binary digits, or more briefly *bits*, a word suggested by J. W. Tukey. A device with two stable positions, such as a relay or a flip-flop circuit, can store one bit of information. N such devices can store N bits, since the total number of possible states is 2^N and $\log_2 2^N = N$. If the base 10 is used the units may be called decimal digits.

- **Paper also chooses to specifically measure information in terms of binary digits**
- **First use of the word “bit”, which is now ubiquitous (*e.g.* smartphone download speeds quoted in Mbit/s, *etc.*)**
- **Binary is highly practical for digital electronic circuitry**

Can you count in binary?

$N(t)$ = the number of signals that can form in time t

The choice of a logarithmic base corresponds to the choice of a unit for measuring information. If the base 2 is used the resulting units may be called binary digits, or more briefly *bits*, a word suggested by J. W. Tukey. A device with two stable positions, such as a relay or a flip-flop circuit, can store one bit of information. N such devices can store N bits, since the total number of possible states is 2^N and $\log_2 2^N = N$. If the base 10 is used the units may be called decimal digits.

$$N(t) = x^t = 2^{t \log x}$$

$$C = \frac{\log_2 N(t)}{t} = \frac{\log_2 x^t}{t} = \log_2 x$$

**channel capacity
(max transmission rate)**

Information rate of Morse code

$$C = \log_2 1.453 = 0.539 \text{ bits/time}$$

Information content

- **How much information is coded by the information source?**
- **The distribution of letters in English is not uniform: “E” is more common than “Q”, “TH” is more common than “QZ”**
- **This means that the information source (if the source is speaking words of English) does not just choose random sequences of dots and dashes**

Acronyms

<u>2moro</u> - Tomorrow	<u>NUB</u> - New person to a site or game
<u>2nite</u> - Tonight	<u>OIC</u> - Oh, I See
<u>BRB</u> - Be Right Back	<u>OMG</u> - Oh My God
<u>BTW</u> - By The Way -or- Bring The Wheelchair	<u>OT</u> - Off Topic
<u>B4N</u> - Bye For Now	<u>POV</u> - Point Of View
<u>BCNU</u> - Be Seeing You	<u>RBTL</u> - Read Between The Lines
<u>BFF</u> - Best Friends Forever -or- Best Friend's Funeral	<u>ROTFLMAO</u> - Rolling On The Floor Laughing My Ass Off
<u>CYA</u> - Cover Your Ass -or- See Ya	<u>RT</u> - Real Time -or- ReTweet
<u>DBEYR</u> - Don't Believe Everything You Read	<u>THX or TX or THKS</u> - Thanks
<u>DILLIGAS</u> - Do I Look Like I Give A Sh**	<u>SH</u> - Sh** Happens
<u>FUD</u> - Fear, Uncertainty, and Disinformation	<u>SITD</u> - Still In The Dark
<u>FWIW</u> - For What It's Worth -or- Forgot Where I Was	<u>SOL</u> - Sh** Out of Luck -or- Sooner Or Later
<u>GR8</u> - Great	<u>STBY</u> - Sucks To Be You
<u>ILY</u> - I Love You	<u>SWAK</u> - Sealed (or Sent) With A Kiss
<u>IMHO</u> - In My Humble Opinion	<u>TFH</u> - Thread From Hell
<u>IRL</u> - In Real Life	<u>RTM or RTFM</u> - Read The Manual -or- Read The F***ing Manual
<u>ISO</u> - In Search Of	<u>TLC</u> - Tender Loving Care
<u>J/K</u> - Just Kidding	<u>TMI</u> - Too Much Information
<u>L8R</u> - Later	<u>TTYL</u> - Talk To You Later -or- Type To You Later
<u>LMAO</u> - Laughing My Ass Off	<u>TYVM</u> - Thank You Very Much
<u>LOL</u> - Laughing Out Loud -or- Lots Of Love -or- Living On Lipitor	<u>VBG</u> - Very Big Grin
<u>LYLAS</u> - Love You Like A Sister	<u>WEG</u> - Wicked Evil Grin
<u>MHOTY</u> - My Hat's Off To You	<u>WTF</u> - What The F***
<u>NIMBY</u> - Not In My Back Yard	<u>WYWH</u> - Wish You Were Here
<u>NP</u> - No Problem -or- Nosy Parents	<u>XOXO</u> - Hugs and Kisses

Partial information quiz

To see a world in a grain of sand

The five boxing wizards jump quickly

H_r_a_d_n_v_r_i_y

W_l_d_n_i_b_t_e_t_a_w_l_s_i_

Partial information quiz

To see a world in a grain of sand

William Blake

The five boxing wizards jump quickly

Pangram

Harvard University

Well done is better than well said

*Benjamin
Franklin*

Shannon's key idea

- **A discrete information source is characterized by the probabilities for emitting signals from the source**

To give a visual idea of how this series of processes approaches a language, typical sequences in the approximations to English have been constructed and are given below. In all cases we have assumed a 27-symbol “alphabet,” the 26 letters and a space.

1. Zero-order approximation (symbols independent and equiprobable).

XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGHYD QPAAMKBZAACIBZL-
HJQD.

2. First-order approximation (symbols independent but with frequencies of English text).

OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI ALHENHTTPA OOBTTVA
NAH BRL.

3. Second-order approximation (digram structure as in English).

ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-
COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

4. Third-order approximation (trigram structure as in English).

IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONS-
TURES OF THE REPTAGIN IS REGOACTIONA OF CRE.

5. First-order word approximation. Rather than continue with tetragram, $\dots$, n -gram structure it is easier and better to jump at this point to word units. Here words are chosen independently but with their appropriate frequencies.

REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NAT-
URAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES
THE LINE MESSAGE HAD BE THESE.

6. Second-order word approximation. The word transition probabilities are correct but no further structure is included.

THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHAR-
ACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT
THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED.

How to simulate the digram structure

3. Second-order approximation (digram structure as in English).

ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-
COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

To construct (3) for example, one opens a book at random and selects a letter at random on the page. This letter is recorded. The book is then opened to another page and one reads until this letter is encountered. The succeeding letter is then recorded. Turning to another page this second letter is searched for and the succeeding letter recorded, etc. A similar process was used for (4), (5) and (6). It would be interesting if further approximations could be constructed, but the labor involved becomes enormous at the next stage.

Measuring information content

Suppose we have syllables with probabilities $p_1, p_2, \dots, p_N$

How much choice do we have in the selection a word?

How uncertain are we of the outcome?

Entropy – measure of disorder or uncertainty in a system

Maximum entropy example

toss a fair coin many times; no way to predict outcome of next toss

Minimum entropy example

toss a two-tailed coin many times; outcome is certain, next toss is tails

Measuring information content

Shannon Entropy

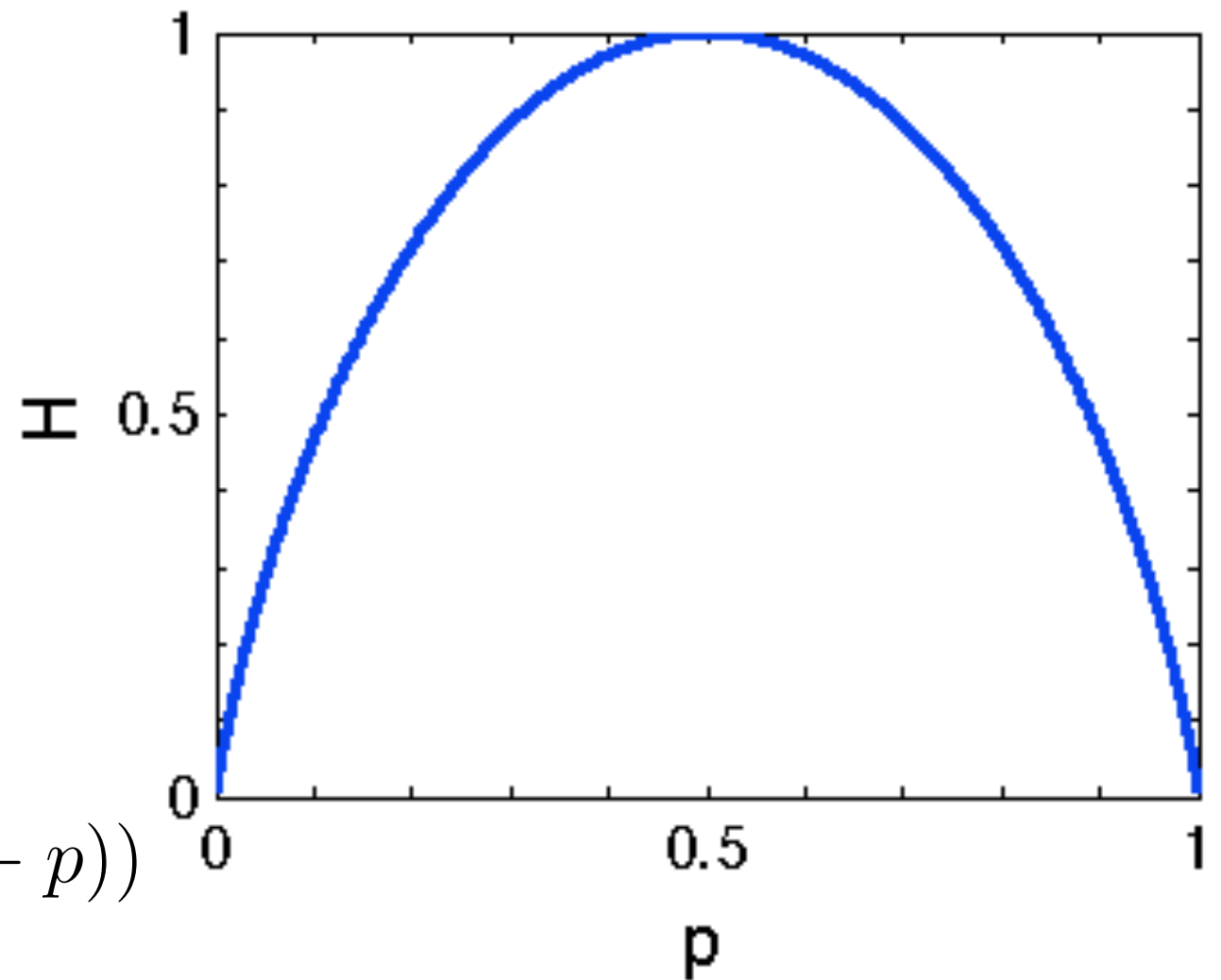
$$H = - \sum_{i=1}^N p_i \log(p_i)$$

Example ($N = 2$)

$$p_1 = p$$

$$p_2 = 1 - p_1 = 1 - p$$

$$H = - (p \log p + (1 - p) \log(1 - p))$$



$H = 0$ only if $p = 0$ or $p = 1$. Does this make sense?

H attains its maximum value at $p = 1/2$. Does this make sense?

Measuring information content

Shannon Entropy

$$H = - \sum_{i=1}^N p_i \log(p_i)$$

Measures the amount of information transmitted

**When you utter a word, how many other choices of words
are you choosing between?**

Measuring information content

What is the probability of the MOST probable word of N symbols?

Suppose we have a word with N symbols.

On average, there will be

$p_1 N$ **occurrences of the first symbol**

$p_2 N$ **occurrences of the second second symbol**

⋮

Hence the probability of the word is

$$P = p_1^{p_1 N} p_2^{p_2 N} \cdots p_N^{p_N N}$$

Measuring information content

The probability of the word is

$$P = p_1^{p_1 N} p_2^{p_2 N} \cdots p_N^{p_N N}$$

In terms of Shannon Entropy

$$P = 2^{-HN}$$

$$H = - \sum_{i=1}^N p_i \log(p_i)$$

How many “most probable” words are there? About $\frac{1}{P}$

Shannon's fundamental theorem

9. THE FUNDAMENTAL THEOREM FOR A NOISELESS CHANNEL

We will now justify our interpretation of H as the rate of generating information by proving that H determines the channel capacity required with most efficient coding.

Theorem 9: Let a source have entropy H (bits per symbol) and a channel have a capacity C (bits per second). Then it is possible to encode the output of the source in such a way as to transmit at the average rate $\frac{C}{H} - \epsilon$ symbols per second over the channel where ϵ is arbitrarily small. It is not possible to transmit at an average rate greater than $\frac{C}{H}$.

- **Suppose the information source has entropy H and the channel has capacity C**
- **One can encode the output of the source so that it can be transmitted a rate of up to C/H symbols per second**
- **It is not possible to transmit faster than C/H !!**

Coming up

Thursday: Authorship – Who wrote the Federalist papers?

Lab: Measure information content in a text

- 1. Determine frequency distribution of a words in a text**
- 2. Check that the distribution obeys Zipf's law (frequency of a word is inversely proportional to its rank in the frequency table)**
- 3. Compute the entropy of the frequency distribution**